

ONLINE MASTERS IN **DATA SCIENCE**

DSC 257R - UNSUPERVISED LEARNING

APPLICATION TO TOPIC MODELING

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Topic Modeling

Given a collection of documents:

- Find an underlying set of topics
- Express each document as a weighted combination of these topics

Example [Blei 2012]

Topics

gene 0.04
dna 0.02
genetic 0.01
...

life 0.02
evolve 0.01
organism 0.01
...

brain 0.04
neuron 0.02
nerve 0.01
...

data 0.02
number 0.02
computer 0.01
...

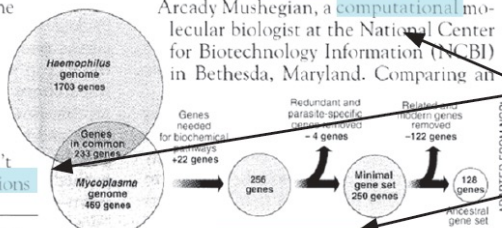
Documents

Seeking Life's Bare (Genetic) Necessities

COLD SPRING HARBOR, NEW YORK—How many **genes** does an **organism** need to **survive**? Last week at the genome meeting here,* two genome researchers with radically different approaches presented complementary views of the basic genes needed for **life**. One research team, using **computer** analyses to compare known **genomes**, concluded that today's **organisms** can be sustained with just 250 genes, and that the earliest life forms required a mere 128 **genes**. The other researcher mapped genes in a simple parasite and estimated that for this organism, 800 genes are plenty to do the job—but that anything short of 100 wouldn't be enough.

Although the numbers don't match precisely, those **predictions**

"are not all that far apart," especially in comparison to the 75,000 **genes** in the human genome, notes Siv Andersson of Uppsala University in Sweden, who arrived at the 800 number. But coming up with a consensus answer may be more than just a **genetic numbers game**, particularly as more and more **genomes** are completely mapped and sequenced. "It may be a way of organizing any newly **sequenced genome**," explains Arcady Mushegian, a **computational** molecular biologist at the National Center for Biotechnology Information (NCBI) in Bethesda, Maryland. Comparing an

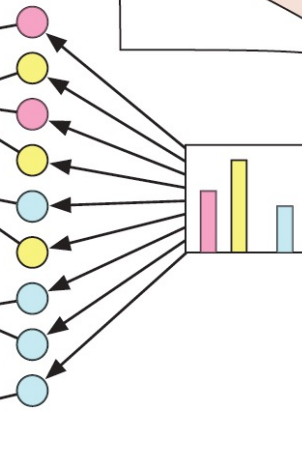


* Genome Mapping and Sequencing, Cold Spring Harbor, New York, May 8 to 12.

Stripping down. Computer analysis yields an estimate of the minimum modern and ancient genomes.

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Topic proportions and assignments



Latent Semantic Indexing (LSI)

Given a large corpus of n documents:

- Fix a vocabulary, say of V words.
- Bag-of-words representation for documents: each document becomes a vector of length V , with one coordinate per word.
- The corpus is an $n \times V$ matrix, one row per document.

	<i>cat</i>	<i>dog</i>	<i>house</i>	<i>boat</i>	<i>garden</i>	<i>...</i>
Doc 1	4	1	1	0	2	
Doc 2	0	0	3	1	0	
Doc 3	0	1	3	0	0	
		$\vdots$				

Let's find a concise approximation to this matrix M .

Latent Semantic Indexing, Cont'd

Use SVD to get an approximation to M : for small k ,

$$\underbrace{\begin{pmatrix} \leftarrow \text{doc 1} \rightarrow \\ \leftarrow \text{doc 2} \rightarrow \\ \leftarrow \text{doc 3} \rightarrow \\ \vdots \\ \leftarrow \text{doc } n \rightarrow \end{pmatrix}}_{n \times V \text{ matrix } M} \approx \underbrace{\begin{pmatrix} \leftarrow \theta_1 \rightarrow \\ \leftarrow \theta_2 \rightarrow \\ \leftarrow \theta_3 \rightarrow \\ \vdots \\ \leftarrow \theta_n \rightarrow \end{pmatrix}}_{n \times k \text{ matrix } \Theta} \underbrace{\begin{pmatrix} \leftarrow \psi_1 \rightarrow \\ \leftarrow \vdots \rightarrow \\ \leftarrow \psi_k \rightarrow \end{pmatrix}}_{k \times V \text{ matrix } \psi}$$

Latent Semantic Indexing, Cont'd

$$\underbrace{\begin{pmatrix} \longleftrightarrow \text{doc 1} \longrightarrow \\ \longleftrightarrow \text{doc 2} \longrightarrow \\ \longleftrightarrow \text{doc 3} \longrightarrow \\ \vdots \\ \longleftrightarrow \text{doc } n \longrightarrow \end{pmatrix}}_{n \times V \text{ matrix } M} \approx \underbrace{\begin{pmatrix} \longleftrightarrow \theta_1 \longrightarrow \\ \longleftrightarrow \theta_2 \longrightarrow \\ \longleftrightarrow \theta_3 \longrightarrow \\ \vdots \\ \longleftrightarrow \theta_n \longrightarrow \end{pmatrix}}_{n \times k \text{ matrix } \Theta} \underbrace{\begin{pmatrix} \longleftrightarrow \psi_1 \longrightarrow \\ \longleftrightarrow \vdots \longrightarrow \\ \longleftrightarrow \psi_k \longrightarrow \end{pmatrix}}_{k \times V \text{ matrix } \Psi}$$

Think of this as a topic model with k topics.

- ψ_j is a vector of length V describing topic j : coefficient ψ_{jw} is large if word w appears often in that topic.
- Each document is a combination of topics: θ_{ij} = weight of topic j in doc i .

Document i originally represented by i th row of M , a vector in $\mathbb{R}^V$.
Can instead use $\theta_i \in \mathbb{R}^k$, a more concise "semantic" representation.